

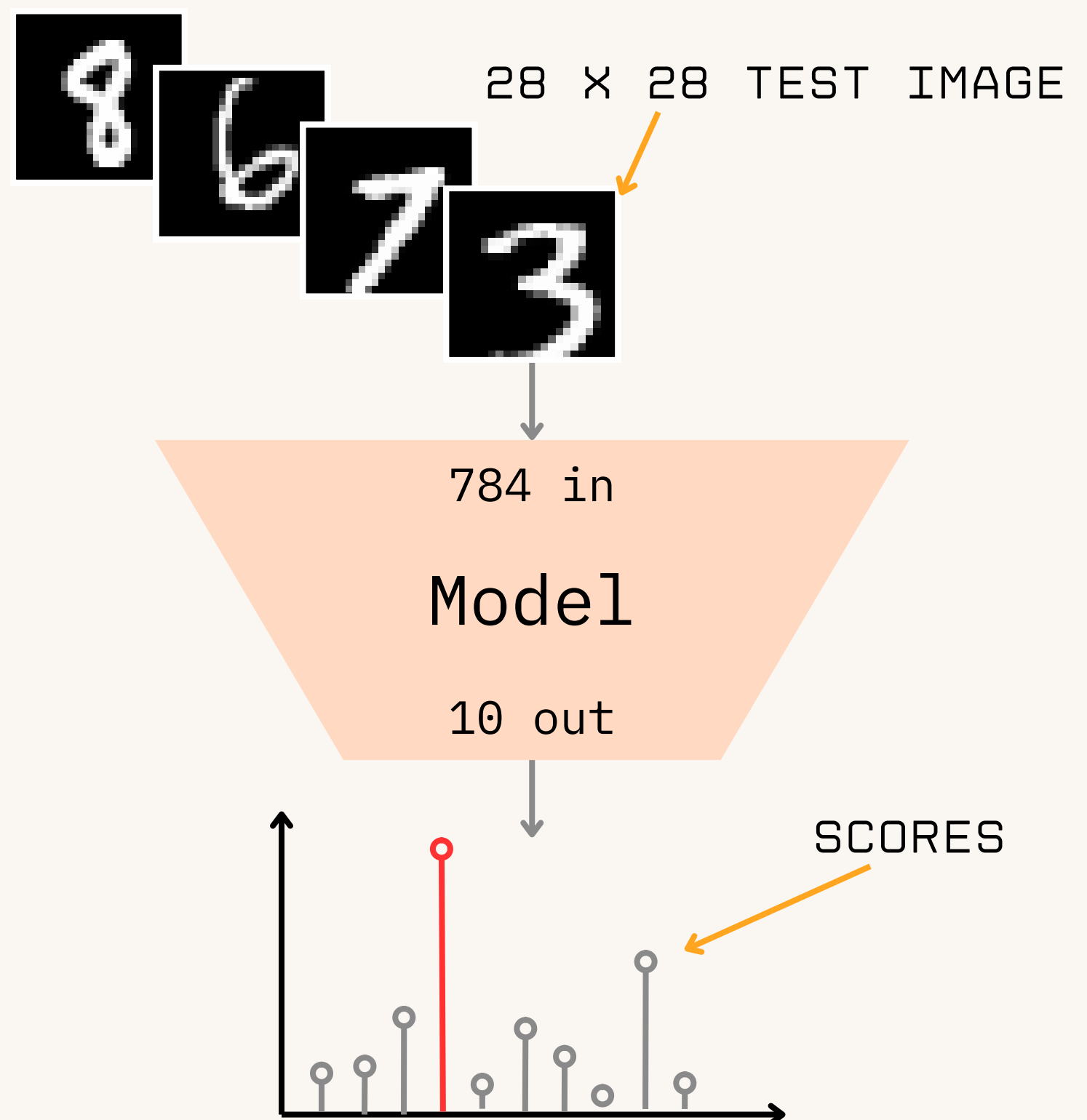
8-BIT ML

Neural Network

Can we run a neural net for real-time digit recognition on vintage hardware?

ALGORITHM

- Train a perceptron (single-layer network) on the MNIST dataset (offline)
- **Quantize** weights to 8-bit integers
- Run inference on 28x28 binarized input images
- 784 integer multiplies per digit, 7840 per decision



THINGS TO TRY

- Sketch a digit on the Koalapad (hold down left button while drawing)
- If the model guesses wrong, how close was it? (check bar graph for confidence levels)

IMPLEMENTATION

- C and Assembly on Atari 800XL (~1983)
- CPU: 6502C @ 1.79 MHz
- RAM: 64 kB

